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W3-02: Hadoop Additional Concepts

COMPSCI4064 (H) and COMPSCI5088 (M)

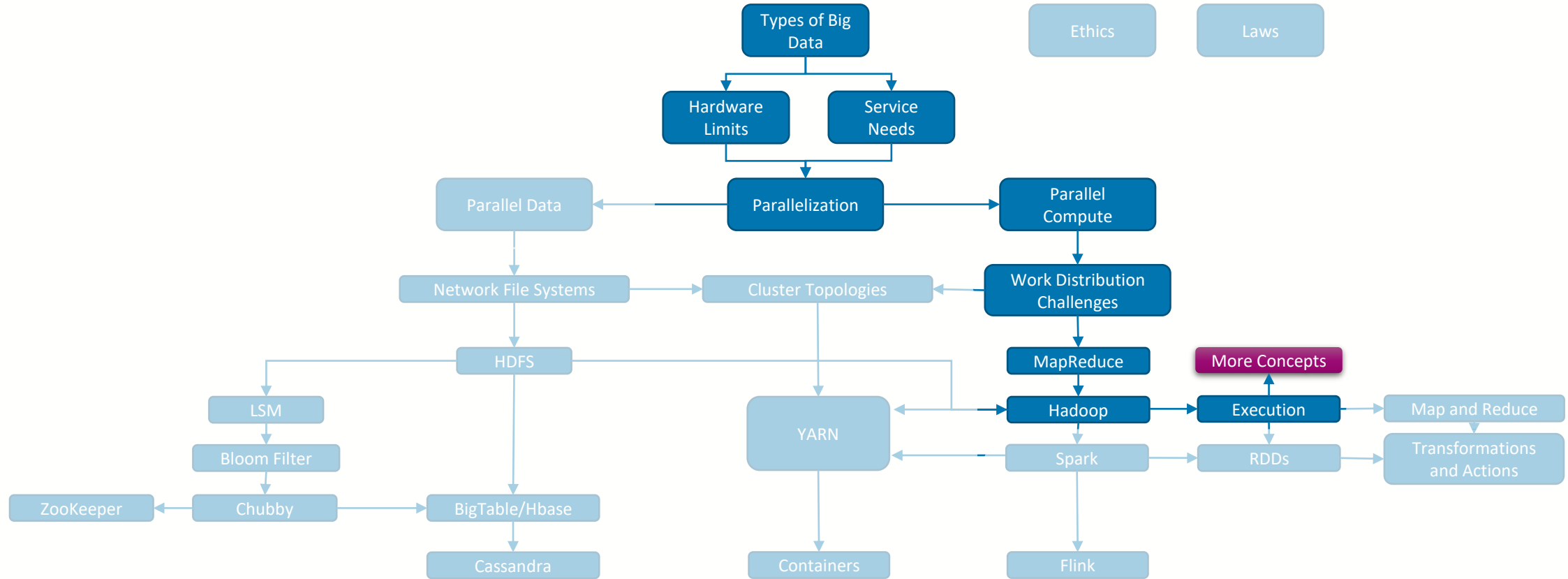
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Where are we?



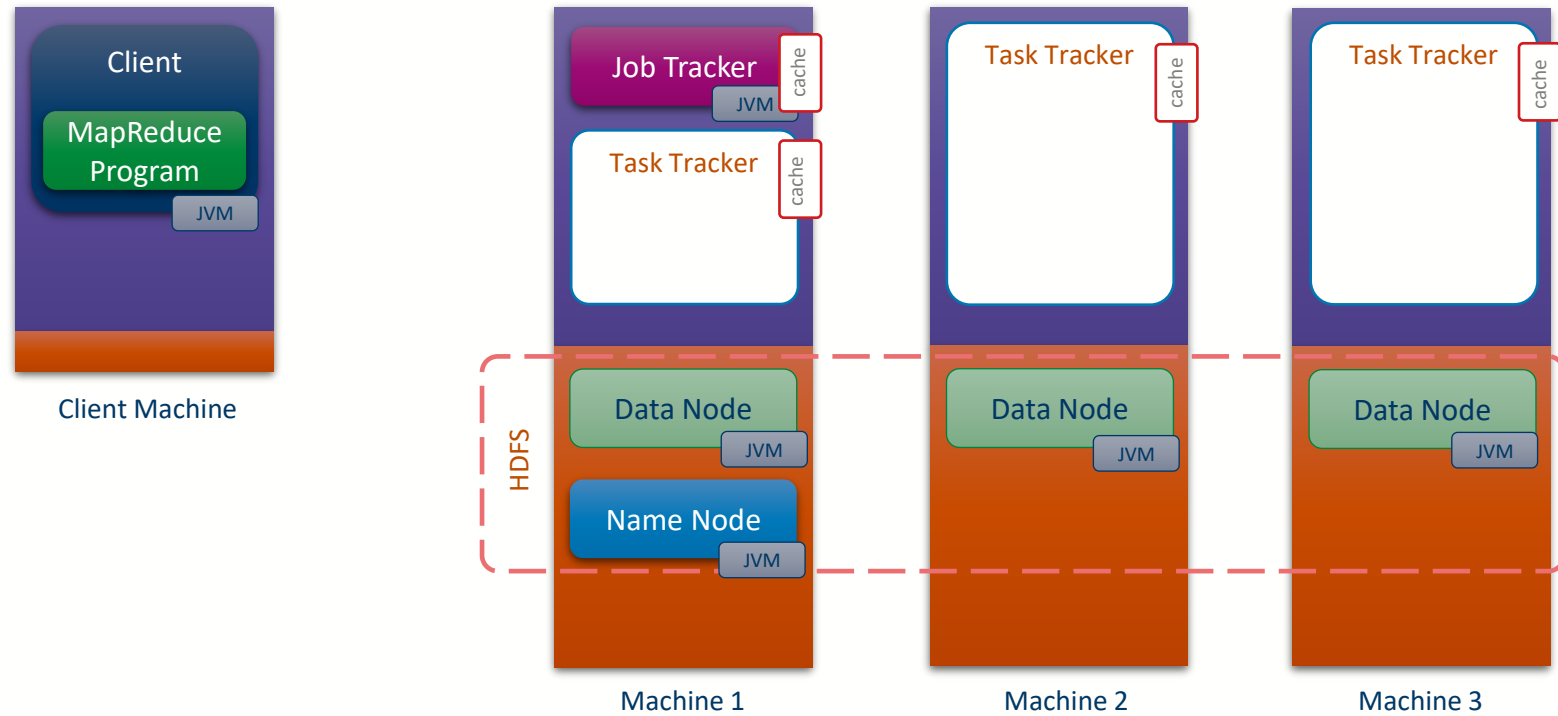


Distributed Cache

- One of the issues with the basic setup of Hadoop is that it requires the **HDFS to store all files**
 - This is an issue for small files, as the HDFS can only read full blocks at a time
- To solve this, Hadoop supports a **distributed cache** as an alternative method to **share files between the Job Tracker and Task Trackers**
 - Is Read-Only
 - Can be used for jars, text, archives, conf files, etc
 - Compressed files unarchived automatically
 - Jars/libs (can be) added to the task's classpath
 - Files are copied only once per job to each tasktracker
 - Can be private or public
 - Can be used as a 'hack' to share state across tasks



Distributed Cache



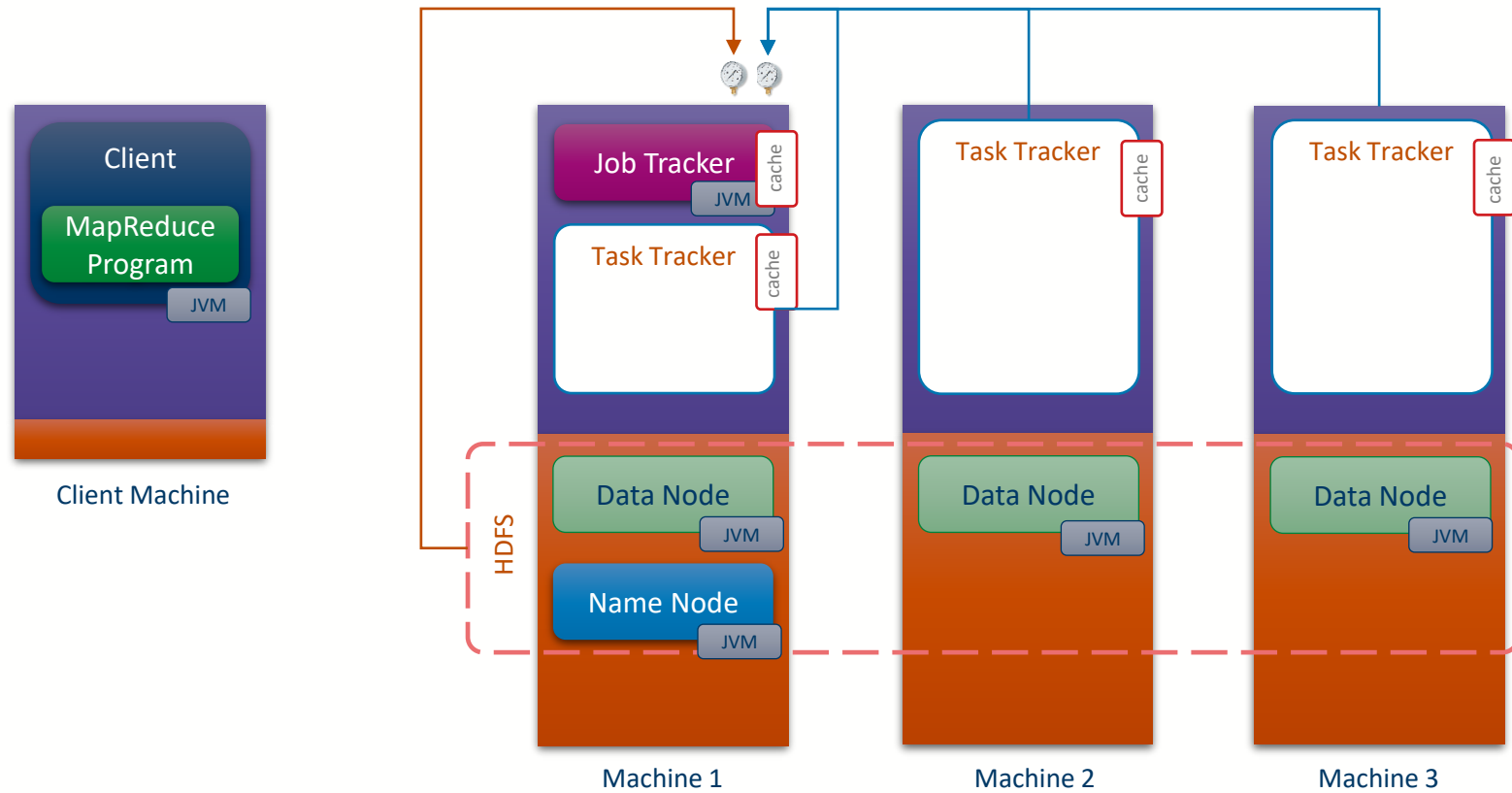


Counters

- While tasks are running, we often want to track the progress of those tasks
 - Hadoop allow tasks to create and update numerical counters at run-time that are periodically shared with the Job Tracker
- Provides built-in counters for:
 - HDFS and FileInputFormat/FileOutputFormat profiling
 - Map/reduce tasks' progress
 - Job progress
- Also support for user-defined counters



Counters





Compression

- Due to the cost of reading and writing data to the HDFS, it is highly recommended that data is compressed
 - More data per HDFS block/partition → Lower storage/network overhead
 - Less I/Os across the dataset → Higher throughput
- This does incur additional CPU cost for decompression
 - But this is usually worth it
- Many options available: ZLib, GZip, BZip2, LZO, Snappy, LZ4
 - But we need to be aware of the effect this has on splitting
 - Bzip2, LZO and Snappy are splittable by default
 - Other options are available, but not for the faint at heart (e.g., see <https://github.com/kevinweil/hadoop-lzo>)



More on Failure Types

Task Failures

- Task failure **detected by Task Tracker**
 - **Runtime exception**/other user code bug
 - Error reported to Task Tracker by child JVM
 - Child **JVM dies**
 - Task Tracker informed by the OS
 - **Task is hung**
 - Detected through timeouts in reporting
- The Task Tracker **informs the Job Tracker of the failure**
 - The Job Tracker reschedules the failed task (to a different Task Tracker, if possible)
 - Multiple attempts per task
- Jobs with many failed tasks are eventually terminated
- ... or bad input records are marked and skipped

Task Tracker Failure

- Task Tracker failure **detected by Job Tracker**
 - Crashed or slow Task Tracker
 - Via **missing heartbeats**
 - Failing or misconfigured Task Tracker
 - Via **multiple task failures**
- Task Tracker **removed from pool of “live” Task Trackers**
- Tasks initially assigned to the failed Task Tracker are rescheduled
- Failed/blacklisted Task Trackers reconsidered after some time



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